

Glyphs Transcript

Hello everyone, welcome to Bitsplanet Digital. Today we will continue our learning journey for the course Data Visualisation Storytelling. We will be now looking at how the various features of Bokeh library can be used for building advanced graphics and use them for creating interactive applications in our storytelling exercise.

We have already seen how Bokeh creates an interactive line plot in the last class and now we will be extending this learning for creating scatter plots, understanding different visualisations for categorical data, how Bokeh interacts with different data structures, so on. So I am importing Pandas as PD for doing calculations and other functions and then I am also importing Bokeh in the Colab environment so that we can execute the Bokeh functions very easily. And remember, always you have to import the output notebook function from Bokeh so that you can plot the graphics and visualise them.

So we have seen the Bokeh plotting interface in last class and we tried to use the figure function for plotting the line graph. So we will skip this example here and directly take the aspect of setting the figure. So whenever you are trying to plot a figure in Bokeh, you have to first call the figure function and set the parameters of the figure.

Define the figure to a particular variable. So in this case we are saying plot is equal to figure. The figure function is assigned to the variable plot.

So you can simply write plot.line or plot.scatter to plot particular functions. And also you can set the height and width of the plot by using the height and width functions. So here plot is equal to figure height and width 300.

So you are creating the figure within the Jupyter notebook. So it is like creating a canvas for your graphic. So this figure with height and width 300 determines the area in which your graphic will come on which you will be superimposing the glides or graphs.

So as you know the lines or the data representation within the Bokeh environment are called glides. So we will first redo the example that we have done in the last class. We will just plot a line.

So I have defined x is equal to some values, y is equal to some values. I am simply plotting the line function between x and y using plot.line command. So plot is the variable to which a figure function is assigned because it is like asking the command.

The command says that you use the figure function to plot the line function for the variables x and y. So that's how it is working. The moment you call plot.line you are using the figure function to draw a straight line function between x and y and it is

created a live object. So now remember you have to use the show function to display the graph.

So you also have to import the show function from Bokeh plotting and show the plot and you are getting the plot function. So you are getting the image finally or the graph finally. So unlike matplotlib you have to use the show function to definitely get the graphic in the Bokeh environment.

Next let us try to add or improvise the graphic. So we have already imported figure and show. There is no need to import them again.

So let us see what additional features can we add to the particular graphic that we have. So we have the plot is equal to figure. We have defined the height and length.

So we want to add the title as basic line plot and also x axis label as x and y axis label as y. OK. So now imagine to do only this part. I am doing the run selection and again always I have to use the show function to get this basic line plot and labels x and y because x and y are not determined here.

So it is not showing. So now again I am running the selection here. OK.

Then I have to use the plot dot line and also I will uncomment this legend label is equal to temperature line width is equal to 2 and colour is equal to red. I have added all the features that I want. This is simple and as similar to matplotlib.

And then you are getting this. So basic line plot x label y label colour is red line width is 2 legend is temperature. Right.

And you have the interactivity features. OK. If I can zoom on this I can zoom this then do the reset and go back.

But this is an advantage of using the bokeh and you have added a line glyph by simply call in setting the figure function to a variable and calling the line function on that particular variable. Right. And setting the height and width.

Now I can simply also change the height and width. Let us say in this case I want to have more wider diagram. Maybe I plot it to 500 and run this.

So I get more wider diagram. You can set the figure as per the space that you want to display the graphic. Now let us try to see one more example of a line chart.

So we are trying to create it for an aeroplane company where we are trying to understand how many million passengers have taken the flights in last 12 months. So this is the initial data. So I am running this month is equal to list 1 to 12.



So I am trying to say each month corresponds to one number starting from 1 to 12 and passengers in million is represented in the array starting from 35.1 million to 54.6 million. Okay. This is what we have drawn.

So I have defined these two variables. Then I am going to create a data frame out of this. Monthly value DFS is equal to create a data frame using PDE data from Pandas data frame.

Data is coming from passengers in millions. So first column is passengers. So it is coming passengers in millions and index column is months.

So these are the two things we are conducted. And DF dot index name is equal to month. We are renaming the index as month.

Then we are trying to see how the data frame looks like. So we have month is equal to 1 to 5 and passengers is equal to the corresponding values from the array. Okay.

Now we will try to plot a figure which shows the line graph which shows the monthly changes in the passenger traffic. So what are we doing? We are simply copying starting with the code. First we have to set the figure.

So you have to first define a variable that takes the figure function and you have the height and width determined. And also you have added a title domestic airline passengers in 2021. So now I have run the selection.

So a figure is defined. A figure function is defined to the plot variable. And I have given the values of X and Y values also right from the particular DF monthly index DF.

Okay. And X is the number of month. Then DF passengers is the Y value.

I also have defined the X and Y variables. All that I have to do now is use the plot dot line function for X and Y variables with a legend passengers line with these three and line colour is blue. And finally show the plot.

So I'm just running this again. So this is what we got. And you think plot is little big can do two.

Then the line is little thicker. You can do a little lesser width also. So you got month 1 2 3 4 up to 12 months.

You have the passengers and we have them. So legend label is passengers. Okay.

So you can do this. So I can I want to add one more feature from here. So yeah.



So I want to add these two. Okay. I want to add more context to this.

So I'm adding this particular part. X axis label is equal to month 10 comma Y axis underscore label is equal to passengers in millions. I'm just running this all this again.

I have run this function. So I got a clear graph showing the monthly passenger traffic and the volume of the passengers. I want to add the titles here.

So I'm adding X axis label is equal to month and Y axis label is equal to so as soon as I type this. Okay. I just got the suggestion and I have got the values and this typing.

So I got X and Y variables. So always remember to define labels. You have to add it in the figure itself.

Okay. Then you are trying to get the graphic. So now you have we will try to plot a scatterplot.

So it's again same. We will try to first determine the figure. So and we can use different features of markers to enhance our scatterplot.

First we have determined the figure with the variable called P. Then we are trying to pass the data directly into the P function P dot gator. So P is a variable which take it is taking the figure function. So now P represents the blank canvas of width 400 and height 200.

And on that you are adding a scatter function scatterplot starting with X values 1 2 3 4 5 Y values 6 7 8 4 5. Okay. And size is 15. So we just run this life object has been created.

Now all you have to do is show dot P. So once you do this you will get your scatterplot. Right. So you'll get the scatterplot.

Now we can try to change the. Okay. We can change the marker type.

Right. And opacity of the colour. Right.

All this we can try to do. So now let us look at the different features that are available or different customisations that you can do. These are X and Y values which are there.

So I'm running the selection. And let me run the each one of this. Now P variable is the figure again this time 400 width to 200.



I am just for the sake of differentiation I am increasing 400 to 400. I am running the run selection. So now one of the marker types to see what they do.

So let us look at the markers. We are plotting X and Y. So we have different scatterplot. Right.

So you have read alpha and marker is triangle. Okay. So I'm just trying to run the entire set.

So this is what you got. Then we are running for yellow. This is yellow.

You got the yellow. Okay. Then you can also try the pink.

So now you are seeing red. So you have yellow pink one being superimposed on the other. So this is how you are getting the diagrams.

So the graphs on one after another. This is how you are getting different types of markers on this particular graphic. So now let us look at the scatterplot using the square pin on the same graph.

So you can see how the markers are changing by using different features. You can improvise the elements on your graphic. Now let us look at adding a line and adding a scatterplot and a line plot together.

Okay. So you want to show them on the same graphic. So combining glyphs into a single plot.

So let us see how a scatterplot and line plot can be combined. So however you can also combine multiple glyphs into a single plot. All this you need to do is to call the multiple glyph functions on the same figure.

Okay. So let us try to show this example. So you have the X and Y variables.

So I am running them to selection. We are plotting the line and plotting the scatter on X and Y. Okay. And then finally trying to see how the graphic will look like.

So you are having the line plot on top of that your scatterplot is also connected. Okay. So now let us try to understand how to represent categorical data in bar charts.

Okay. Bar charts represent data for different categories as vertical or horizontal bars. So we can use the continuous X axis range or we can also use it as a categorical data.

In last class we have seen in Seaborn example how tips data set where you had the total bill amount versus the tip amount. How the different categorical variables like

gender, the day, the time of visiting the restaurant. Smoker and non-smoker type have been used to bring out interesting features using different types of graphs.

And bar chart was one of those use cases. So now let us try to understand how categories can be used instead of continuous numbers. And then how to pass these commands into the syntax so that you can generate graphs, bar graphs, which can compare the categorical variables.

Let us first start with a data set which compares the number of fruits that you have. We have apples, pears, plums, grapes, nectarines and strawberries. And you have the counts of each of these.

Each of these is a category. Apple is a category, pear is a category and there are counts. There are two variables, fruit and counts, counts of fruit.

But each count, each fruit has, I mean, within the fruits there are different types of fruits. Therefore they form the categorical variable. Now I am plotting, calling again the figure function.

First of all we need to define the figure function in Bokeh before we plot the figure. So `p` is equal to figure, `x` underscore range is equal to fruits. We are calling the `x` underscore range function to show it is a discrete variable.

And then setting the height to 350, title is fruit counts. By calling the `x` range function, we are able to tell that to use categories instead of continuous range of numbers for the Bokeh. So we are telling the system to use the discrete range by specifying `x` range is equal to a particular array.

And they have to be in a particular order in which the graphic will be produced. So we have set this now. Now we are actually passing the final data on how to get a bar graph.

So we are plotting `p` dot `v` bar, ok. `x` fruits, top counts, ok. Then width, right, this is what we are.

And colour is firebrick and show plot `p`. We got this clearly, fruit counts, ok. So then you have the apples, pears, right, nectarines, plums and strawberries. You can also try to do this.

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